Vishal Naveen Akkala | AI/ML Engineer

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SUMMARY

AI/ML Engineer with 5+ years of experience developing machine learning models, Generative AI applications, and scalable data solutions across financial services, healthcare, and retail domains. Skilled in predictive modeling, Retrieval-Augmented Generation (RAG), NLP, MLOps, model deployment, and large-scale data processing using Python, SQL, PyTorch, Scikit-Learn, Databricks, AWS, and Azure. Experienced in building production-ready systems, optimizing model performance, and delivering AI-driven solutions that improve operational efficiency, decision-making, and customer outcomes.

PROFESSIONAL EXPERIENCE

AI/ML Engineer — **Apr 2025 - Present | FL, USA**
Capital One

- Built Retrieval-Augmented Generation (RAG) applications using LangChain, vector databases, embeddings, and large language models, improving information discovery across internal knowledge repositories and reducing research effort for business users.
- Developed classification, forecasting, and anomaly detection models using Python, Scikit-Learn, XGBoost, and PyTorch, enabling data-driven decision-making and improving the accuracy of operational and analytical processes.
- Processed more than 15 million records monthly through PySpark and Databricks pipelines, accelerating feature engineering workflows and supporting scalable machine learning development across multiple business initiatives.
- Integrated AI-powered capabilities into enterprise applications through REST APIs, enabling semantic search, document summarization, and conversational assistance functionalities used by cross-functional business teams.
- Fine-tuned transformer-based models using domain-specific datasets to strengthen document classification, information extraction, and search relevance across large collections of structured and unstructured content.
- Automated model deployment workflows using MLflow, Docker, GitHub Actions, and cloud services, shortening release cycles while improving consistency, traceability, and operational reliability.
- Implemented monitoring solutions tracking prediction quality, feature drift, model health, and data anomalies, supporting stable machine learning performance across production environments.
- Designed reusable feature engineering frameworks using SQL, Python, and Databricks, reducing preparation effort and improving training dataset quality for machine learning applications.
- Collaborated with software engineers, analysts, and product stakeholders to translate business requirements into practical AI solutions, increasing adoption of machine learning capabilities across operational workflows.
- Evaluated model performance through experimentation, hyperparameter optimization, A/B testing, and error analysis, improving model effectiveness while maintaining scalability and maintainability standards.

ML Engineer — **Feb 2020 - Dec 2023 | India**
Accenture

- Developed machine learning solutions for financial services, healthcare, and retail clients, supporting predictive analytics initiatives that improved reporting accuracy, operational visibility, and business decision-making capabilities.
- Constructed end-to-end machine learning pipelines covering data ingestion, transformation, feature engineering, model training, validation, deployment, and monitoring across multiple client engagements.
- Built regression, classification, and forecasting models using Python, SQL, Scikit-Learn, and XGBoost, helping stakeholders identify business trends, risks, and performance improvement opportunities.
- Performed exploratory data analysis on datasets exceeding 10 million records, uncovering actionable patterns, improving data quality, and supporting more effective model development activities.
- Delivered natural language processing solutions for sentiment analysis, document categorization, and text classification, reducing manual processing effort and improving operational efficiency.
- Automated data preparation, validation, and model training workflows using Python and Apache Airflow, improving processing reliability and accelerating analytical project delivery timelines.
- Developed recommendation and predictive analytics models that strengthened customer engagement strategies and supported personalized user experiences across digital platforms.
- Partnered with cloud engineering teams to migrate machine learning workloads to AWS and Azure environments, improving scalability, accessibility, and operational efficiency.

- Created Power BI and Tableau dashboards presenting model outputs, performance metrics, and business insights, enabling stakeholders to monitor outcomes and make informed decisions.
- Supported model validation, retraining, experiment tracking, and performance monitoring activities, ensuring deployed machine learning solutions continued delivering reliable and measurable business value.

EDUCATION

Master of Science in Information Systems and Operations Management **Dec 2025 | FL, USA**
University of Florida
B.Tech. in Electrical and Electronics Engineering **Jun 2021 | India**
National Institute of Technology

TECHNICAL SKILLS

Programming Languages: Python, SQL, PySpark, R
Machine Learning: Classification, Regression, Forecasting, Anomaly Detection, Recommendation Systems, Predictive Modeling, Feature Engineering, Model Evaluation, Hyperparameter Optimization, Statistical Analysis
Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, Embeddings, Semantic Search, Vector Databases, LLM Evaluation, Fine-Tuning
Natural Language Processing: Transformer Models, Hugging Face, Text Classification, Information Extraction, Sentiment Analysis, Named Entity Recognition (NER)
AI/ML Frameworks: Scikit-Learn, PyTorch, TensorFlow, XGBoost, Pandas, NumPy
MLOps & Deployment: MLflow, Docker, CI/CD Pipelines, GitHub Actions, Model Deployment, Model Monitoring, Experiment Tracking, A/B Testing, Model Validation
Cloud & Data Platforms: AWS, Azure, Azure OpenAI, AWS SageMaker, Databricks
Data Engineering: Apache Spark, Apache Airflow, ETL Pipelines, Data Processing, Data Transformation, Data Modeling
Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Pinecone, ChromaDB
APIs & Integration: REST APIs, Microservices, API Integration
Visualization: Power BI, Tableau
Tools: Git, GitHub, Jupyter Notebook, Linux

PROJECTS

Enterprise Knowledge Assistant using RAG and LLMs
Technologies: Python, LangChain, OpenAI, Pinecone, FastAPI, Docker
- Developed a Retrieval-Augmented Generation application leveraging LangChain, OpenAI models, and Pinecone vector search to deliver context-aware responses across enterprise knowledge repositories.
- Implemented document ingestion, chunking, embeddings generation, and retrieval pipelines, improving answer relevance while enabling efficient semantic search across unstructured content.
- Exposed AI capabilities through FastAPI endpoints and Docker containers, supporting scalable deployment, streamlined integration workflows, and consistent inference performance across environments.

Customer Churn Prediction and MLOps Platform
Technologies: Python, XGBoost, MLflow, Airflow, AWS, Docker
- • Built customer churn prediction models using XGBoost and engineered behavioral features, identifying at-risk customers and supporting retention strategies through predictive insights.
- • Automated model training, experiment tracking, version control, and deployment workflows using MLflow and Airflow, reducing manual intervention across machine learning operations.
- • Established monitoring processes for prediction quality, feature drift, and model performance, enabling reliable production operations and data-driven model maintenance activities.